Melody Molander

molander.3@osu.edu
melodymolander.com

POSITIONS

Ross Assistant Professor — Fall 2025 - present
The Ohio State University

Adjunct Professor — Spring 2018 - Spring 2019
Cosumnes River College

EDUCATION

Ph.D. in Mathematics — Fall 2019 - Spring 2025
University of California, Santa Barbara
- Advisor: Stephen Bigelow
- Thesis: *Skein Theory of Affine ADE Subfactor Planar Algebras*
- Certificate in College and University Teaching

M.A. in Mathematics — Fall 2014 - Spring 2017
University of Oklahoma

B.S. in Mathematics — Fall 2010 - Spring 2014
University of California, Davis

PAPERS

7. *Dihedral pointed fusion categories: diagrammatics, 3-cocycle representatives, and planar algebras*
(with A. Czenky)
Submitted. arXiv:2610.40498

6. *Thompson's group V and virtual link theory*
(with M. Chrisman and L. Liles)
Submitted. arXiv:2607.28406

5. *NIM-representations of Tambara-Yamagami generalizations*
(with A. Czenky, E. McGovern, M. Müller, and A. Ros Camacho)
Submitted. arxiv:2602.23502

4. *A mirror deformation of Markov numbers*
(with L. Bittmann, P. Jouteur, E. Kantarcı Oğuz, and E. Yıldırım)
Submitted. arxiv:2602.14802

3. *A well-defined jellyfish algorithm for the affine E_7 subfactor planar algebra*
Journal of Pure and Applied Algebra 230, 11 (2026), Paper No. 108383.

2. *Skein theory of affine A subfactor planar algebras*
Submitted. arxiv:2410.05519

1. *A determinant formula of the Jones polynomial for a family of braids*
(with D. Asaner, S. Kumar, A. Pease, and A. Poudel)
Topology and its Applications 384 (2026), Paper No. 109791, 18.

RESEARCH INTERESTS

Quantum algebra, quantum topology, operator algebras, tensor categories, diagrammatic algebras, subfactors, classical and virtual knot theory, quantum groups, higher categories, low-dimensional topology

SELECTED AWARDS & GRANTS

AMS-Simons Travel Grant	2026-2028
UCSB Doctoral Student Travel Grant	2025
USTARS Distinguished Graduate Student in Topology	2024
Decolvenaere Research Accelerator Award, UCSB Graduate Division	2023
Operator Algebra Mentor Network Grant	2023
Individualized Professional Skills (IPS) Grant, UC Santa Barbara	2023

MENTORSHIP

The Ohio Summer Undergraduate Institute in Mathematics Mentor	May 2026 - August 2026
UCSB Directed Reading Program Mentor	December 2019 - June 2023
REU Graduate Mentor, UCSB	June 2022 - August 2022
Women in STEM Mentor, UCSB	September 2019 - June 2022

TEACHING
***=ONLINE COURSE**

Instructor of Record - The Ohio State University — Fall 2025 - present
Math 5801 - General Topology and Knot Theory (upcoming)
Math 2153 - Calculus 3
*Math 1116 - Excursions in Mathematics
Math 1131 - Calculus for Business

Lead TA - UC Santa Barbara — Fall 2022 - Spring 2023
Provided professional development and training to new math department TAs

Teaching Assistant - UC Santa Barbara — Fall 2019 - Spring 2025
MATH 111ABC - Abstract Algebra
MATH 108AB - Introduction to Linear Algebra
MATH 8 - Transition to Higher Mathematics
*MATH 8 - Transition to Higher Mathematics
*MATH 6AB - Vector Calculus
MATH 34B - Calculus for Social and Life Sciences

Associate Instructor - UC Santa Barbara — Summer 2020, 2021, 2024
MATH 4A - Linear Algebra with Applications
*MATH 6A - Vector Calculus I
*MATH 4A - Linear Algebra with Applications

Adjunct Mathematics Professor - Cosumnes River College — Spring 2018 - Spring 2019
MATH 30 - Pre-Algebra
STAT 300 - Introduction to Statistics and Probability
MATH 20 - Arithmetic

Teaching Assistant - University of Oklahoma — Fall 2015 - Spring 2017
MATH 1743 - Business Calculus I
MATH 1914 - Calculus 1 (Fast Track)
MATH 2924 - Calculus 2 (Fast Track)
MATH 1823 - Calculus 1

Instructor of Record - University of Oklahoma — Fall 2014 - Summer 2015 & 2016
MATH 1823 - Calculus 1
MATH 1503 - College Algebra

SERVICE

AMS 2026 Fall Eastern Sectional Special Session Co-Organizer	March 2026 - October 2026
Operator Algebra Mentor Network Mentee Board Member	September 2025 - August 2026
AMS 2025 Fall Eastern Virtual Sectional Special Session Co-Organizer	February 2025 - October 2025
Topology Seminar Co-Organizer	October 2024 - June 2025
AWM Student Chapter at UCSB, Secretary	June 2023 - June 2025
AWM EvenQuads 3 Data Collector	June 2024
Directed Reading Program at UCSB Co-Organizer	December 2020 - July 2023
Graduate Quantum Algebra and Topology Seminar Co-Organizer, UCSB	July 2023 - January 2024
AWM Chapter at UCSB Founder and President	August 2022 - June 2023
Quantum Algebra and Topology Seminar Co-Organizer, UCSB	September 2021 - June 2022
AWM Essay Contest Judge	January 2020 - March 2020
Math Teacher's Circle - Oklahoma City Founder and Instructional Leader	April 2015 - June 2017

REFEREE

Journal of the Korean Mathematical Society
Theory and Applications of Categories

RESEARCH VISITS

Institute of Mathematics of the Polish Academy of Sciences, — May 2027 - June 2027
The Simons Semester on "Quantum Groups and Operator Algebras"

TALKS *=INVITED

*47. Joint Mathematics Meetings; USTARS at JMM: A Special Session on Algebra and Topology; McCormick Place Convention Center — January 2027

*46. Joint Mathematics Meetings; Advances in Operator Algebras and Noncommutative Geometry; McCormick Place Convention Center — January 2027

*45. AMS Fall Western Sectional Meeting; Recent Developments in Tensor Categories and Their Applications; Arizona State University — November 2026

*44. North Carolina Algebra and Mathematical Physics Conference; University of North Carolina Wilmington	October 2026
43. AMS Fall Eastern Sectional Meeting; Thompson's Groups, Braid Groups, and Their Applications to Knot Theory; George Washington University	October 2026
42. Advances in Geometric and Quantum Topology; Temple University	July 2026
*41. Modern Methods in Representation Theory; University of Ottawa	July 2026
40. Great Plains Operator Theory Symposium; University of Iowa	May 2026
*39. Quantum Symmetries Conference IV; Illinois State University	May 2026
*38. Moab Topology Conference; Utah State University, Moab	May 2026
*37. UNCW Algebra and Number Theory Seminar; University of North Carolina Wilmington	February 2026
*36. Banff International Research Station Workshop; New Developments in Tensor Categories (Virtual)	January 2026
*35. Joint Mathematics Meetings; Recent Progress in Quantum Topology Special Session; Walter E. Washington Convention Center	January 2026
*34. Joint Mathematics Meetings; Recent Developments by Women in Mathematical Physics Special Session; Walter E. Washington Convention Center	January 2026
*33. Banff International Research Station Workshop; Operator Algebraic Quantum Groups	December 2025
*32. West Coast Operator Algebra Seminar; Pomona College	December 2025
*31. European Quantum Algebra Lectures; Cardiff University and University of Nottingham (Virtual)	November 2025
30. Quantum Symmetries Conference III; Iowa State University	November 2025
*29. East Coast Operator Algebras Seminar; University of Nebraska - Lincoln	October 2025
28. Young Mathematicians in C* Algebras; University of Southern Denmark	July 2025
*27. AWM Research Symposium; Non-commutative algebras, tensor categories, and diagrams Special Session; University of Wisconsin-Madison	May 2025
*26. AMS Spring Western Sectional Meeting; Knot Theory and Quantum Topology Special Session; California Polytechnic State University	May 2025
25. Banff International Research Station; Women in Noncommutative Algebra and Representation Theory 4	March 2025
24. Quantum Groups, Tensor Categories and Quantum Field Theory Workshop; University of Oslo	January 2025
*23. Joint Mathematics Meetings; Knots, 3-manifolds, and Their Invariants Special Session; Seattle Convention Center	January 2025
22. Joint Mathematics Meetings; Advances in Operator Algebras Special Session; Seattle Convention Center	January 2025
*21. AMS Western Sectional Meeting; Graphical Calculus in Representation Theory and Low-Dimensional Topology Special Session; University of California, Riverside	October 2024
20. Quantum Groups and Representation Theory; North Carolina State University	October 2024
19. Young Mathematicians in C Algebras; University of Glasgow	August 2024
18. Oberwolfach Seminar; Classification of C-Algebras and Dynamics Seminar; Oberwolfach Research Institute for Mathematics	May 2024
17. Banff International Research Station Workshop; Skew Braces, Braids, and the Yang-Baxter Equation; Banff International Research Station	May 2024
*16. USTARS Distinguished Graduate Student in Topology Speaker; Underrepresented Students in Topology and Algebra Research Symposium; University of Iowa	April 2024

15. Graduate Student Topology and Geometry Conference; Michigan State University	April 2024
14. Southeastern Lie Theory Workshop; Quantum Structures in Lie Theory; University of Virginia	March 2024
13. Women in Mathematics in Southern California Symposium; Pomona College	February 2024
*12. Joint Mathematics Meetings; Knots, Skein Modules, and Categorification Special Session; Moscone Center	January 2024
*11. Claremont Topology Seminar; Pomona College	November 2023
10. Twinned Conference on C-Algebras and Tensor Categories; Fields Institute; and streamed live at International Centre for Mathematical Sciences	November 2023
9. Young Mathematicians in C*-Algebras; KU Leuven	August 2023
8. Great Plains Operator Theory Symposium; The Ohio State University	May 2023
7. AMS Spring Western Sectional; Algebraic Structures in Knot Theory Special Session; Fresno State University	May 2023
6. Moab Topology Conference; Utah State University, Moab	May 2023
5. Redbud Topology Conference; University of Oklahoma	March 2023
4. Underrepresented Students in Topology and Algebra Symposium; University of Washington	March 2023
*3. Quantum Groups Seminar; Université de Lille & University of Oslo (Virtual)	March 2023
2. Tech Topology Conference; Georgia Institute of Technology	December 2022
1. Topology Students Workshop; Georgia Institute of Technology	July 2022

WORKSHOPS

Institute of Science and Technology Austria (ISTA)	
Women in Topology V	January 2027
International Centre for Mathematical Sciences (ICMS)	
Women in Mathematical Physics 3	April 2026
Oberwolfach Research Institute for Mathematics (MFO)	
Classification of C*-Algebras and Dynamics Seminar	May 2024
Banff International Research Station (BIRS)	
Operator Algebraic Quantum Groups	November 2025
Women in Noncommutative Algebra and Representation Theory 4	March 2025
Skew Braces, Braids, and the Yang-Baxter Equation Workshop	May 2024
Women in Operator Algebras 3	June 2023
American Institute of Mathematics (AIM)	
Quantum Invariants and Low-dimensional Topology	August 2023

REFERENCES

7. Noah Snyder, Indiana University Bloomington, nsnyder@gmail.com
6. Micah Chrisman, The Ohio State University, chrisman.76@osu.edu
5. David Penneys, The Ohio State University, penneys.2@osu.edu
4. Stephen Bigelow, University of California, Santa Barbara, bigelow@math.ucsb.edu
3. Zhenghan Wang, Microsoft Station Q and University of California, Santa Barbara, zhenghwa@math.ucsb.edu
2. Ana Ros Camacho, Universitat de València, ana.ros.camacho@uv.es
1. Deborah Stout, The Ohio State University, stout.261@osu.edu (concerns teaching)